Group - B

Assignment No. 11

• **Aim:**

write a code in java for simple word count application that number of occurances of each word in given input set using the hadoop Map-Reduce frameworks on local-standalone set-up.

• **Theory:**

Map-Reduce :

Map-Reduce is a framework using which we can write applications to process huge amounts of data, in parallel, on large clusters of commodity hardware in a reliable manner.

MapReduce is a processing technique that a program model for distributed computing based on java. The MapReduce algorithm contains two important tasks namely Map and Reduce. Map takes a set of data and converts it into another set of data, where individual elements are broken down into tuples secondly, reduce task which takes the output from a map as an input and combines those data tuples into a smaller set of tuples. As a sequence of the name mapReduce implies, the reduce tasks is always performed after the map job.

The Hadoop, MapReduce is a combination that decomposes large manipulation jobs into individual tasks that can be executed in parallel across a cluster of servers. The results of tasks can be joined together to compute final results.

MapReduce consist of 2 steps:

- Map function:
  - It takes a set of data and converts it into another set of data, where individual elements are broken down into tuples (key-value pair).

eg - Map function in word count:

Input : set of Data
like - BUS, Car, bus, car, train, car, bus, car, TRAIN, BUS, bus, CAR, CAR, car

Output : convert into another set of data
(Bus,1), (car,1), (bus,1), (car,1), (train,1), (car,1), (bus,1), (car,1), (train,1), (bus,1), (TRAIN,1), (BUS,1), (bus,1), (caR,1), (car,1).

Reduce function:
- Takes the output from map as an input and combines those data tuples into a smaller set of tuples.

- e.g., Reduce function in word count.

Input :- set of tuples : (Bus,1), (car,1), (bus,1) (car,1).

Output : convert into smaller set of tuples
like :- (BUS, 7), (CAR, 7), (TRAIN, 4)

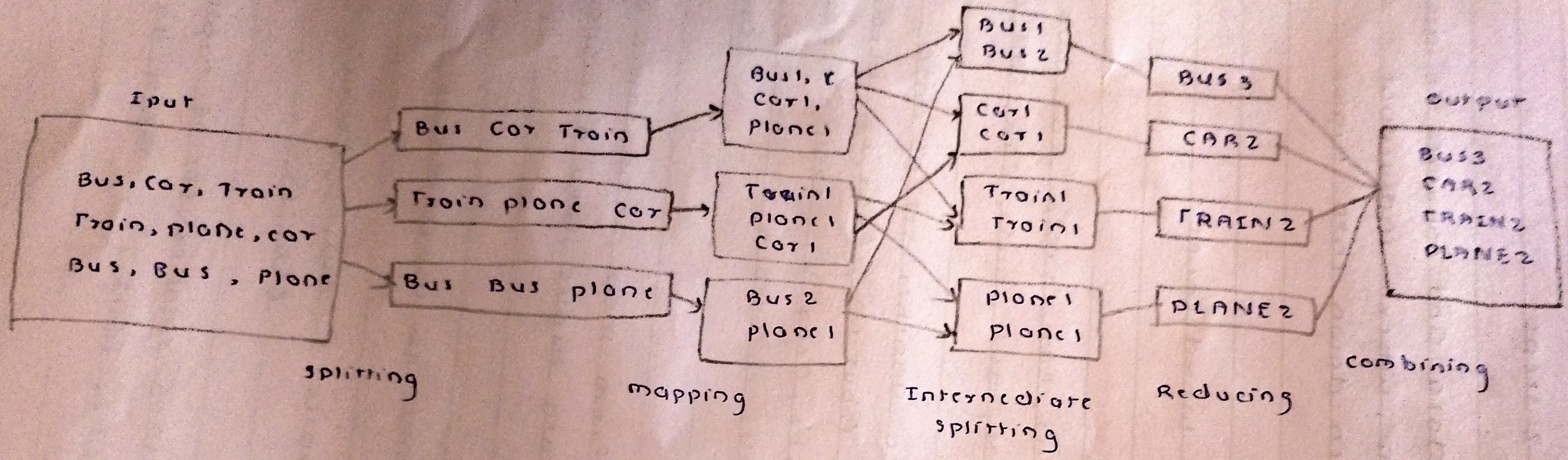

fig. workflow of MapReducing

Workflow of MapReduce consist of 5 steps:

1. Splitting:
   - The splitting parameter can be anything, e.g., splitting by space, comma, semicolon, or even by a new line.

2. Mapping:
   - as explained above.

3. Intermediate splitting:
   - The entire process in parallel on different clusters. In order to group them in "Reduce phase" the similar KEY data should be on the same cluster.

4. Reduce:
   - It is nothing but mostly group by phase

5. Combining:
   - The last phase where all the data is combined together to form a result.

Conclusion:

Implemented code in simple word count application that counts the number of occurances of each word in a given input set using Hadoop Map-Reduce.

Assignment No. 12

Aim:

Locate dataset (e.g. sample_wheather.txt) for working on wheather data which reads the text input files and finds average for temperature dew point and wind speed

Theory:

Wheather and climate data are valuable resources not only to forecast the weather but for dozens of purpose across industries to apply data analytics and data science technique to these data to inform and support decision-making are almost endless.

We write a program for analyzing wheather datasets to understand its data processing programming model. Weather sensors are collecting wheather information across the globe in a large volume of log data. This wheather data is semi-structured and record oriented.

This data is stored in a line-oriented ASCII format, where each row represents a single record. Each row has lots of fields like longitude, latitude daily max-min temperature, daily average temp. etc. for easiness, we will focus on the main element i.e. temperature. We will use the data from the national centres for environmental information (NCEI). It is a massive amount of historical wheather data that we can use for our data analysis.

Data processing steps:

The program follows these logical steps:

1. Read Input file:
   - The dataset file is opened and all records are accessed

2. Extract Required fields:
   - From each line, values of temperature, dew point and wind speed are extracted.

3. Convert Data types:
   - Since file data is read as text, it must be converted into numerical form (e.g. float or integer).

4. Accumulate Values:
   - Separate variables are used to store the sum of:
     - temperature
     - Dew point
     - wind speed

5. Count records:
   - Keep track of the number of entries in the dataset.

Average Calculation:

The average is calculated using the standard formula:

$$\text{Average} = \frac{\text{Sum of all values}}{\text{Number of records}}$$

This formula is applied separately for:

- Average temperature
- Average Dew point
- Average wind speed

- Conclusion:

We have successfully calculated average of temperature, dew point and wind speed over temperature data.

Assignment No. 13

Aim:

write a simple program in SCALA using Apache spark framework.

Theory:

what is SCALA?

Scala is an acronym for "Scalable language". It is a general-purpose programming language designed for programmers who want to write programs in a concise, elegant and type-safe way. Scala enables programmers to be more productive. Scala is developed as an object oriented and function programming language.

If you write a code in scala, you will see that the style is similar to a scripting language. Even through scala is a new language, it has gained enough users and has a wide community support. It is one of the most user-friendly languages.

Scala is pure object oriented programming language:

Scala is an object-oriented programming language everything in scala, allow you to add new operations to existing classes with the help of implicit classes. One of the advantage of scala is that it makes it very easy to interact with java code. You can also write a java code inside scala class.

SCALA is a functional language:

SCALA is a programming language that has implemented major functional programming concepts in functional programming, every computation is treated as a mathematical function which avoids states and mutable data. The functional programming every computation is treated exhibits following characteristics:

- proper and flexibility
- simplicity
- suitable for parallel processing

Installing SCALA:

SCALA can be installed in any unix or windows based systems, below are steps to install for ubuntu (14.04) for scala version 2.11.7.1 am showing the steps for installing scala (2.11.7) with java version 7. It is necessary to install java before installing scala. you can also install latest version of scala (2.12.1) as well.

step 0: open the terminal

step 1: Install java

```
$ sudo apt add-repository ppa: webupd8
     team / java
$ sudo apt-get update
$ sudo apt-get install oracle-java7-
                                installer
$ java -version
```

step 2: once java is installed we need to install scala.

$ cd ~/Downloads

$ wget http://www.scala-lang.org/files/archieve/scala-2.11.7.deb.

$ sudo dpkg -i scala-2.11.7.deb

$ scala -version

SCALA Basic terms:

- object:
  - An entity that has state and behavior is known as an object. for example: table, person, car etc.

- class:
  - A class can be defined as a blueprint or template for creating different objects which defines its properties and behavior.

- Method:
  - It is a behavior of a class, A class can contain one or more than one method. for example deposite can be considered as a method of bank class.

- closure:
  - closure is any function that closes over the environment in which its defined. A closure returns value depends on the value of one or more variables which is declared outside this closure.

- Traits:
  - Traits are used to define object types by specifying the signature of the supported methods. It is like interface in java.

- Conclusion:

  We have installed and executed scala program successfully.